

Economics of Sanctions

Part 2 — Firm-level Adjustment

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Why focus on firms?

- Macro counterfactuals (Part 3) tell us **aggregate welfare**: Russia -2.6% , EU -0.1% in 2022 (Crozet et al., 2025).
- But the **adjustment is heterogeneous**: a few hundred firms drive most of the lost trade, and the firms that exit do not come back.
- Granular customs data let us see **margins of adjustment, compliance vs. evasion**, and **collateral damage** that the GE model averages away.

Source: IATRC keynote frame 2; Crozet et al. (2025) for the macro figures.

Roadmap — six blocks, 25 frames

1. Sanctioning side — **extensive margin**: who exits? (Crozet et al., 2021)
2. **Estimation & gravity block** — PPML, LPM, triple-diff (5 frames, contiguous).
3. Sanctioning side — **intensive margin**: Friendly Fire (Crozet and Hinz, 2020).
4. Target side — AHÖ, Nigmatulina, Miromanova, Görg et al.
5. Frontier — circumvention (Scheckenhofer et al., 2025; Chupilkin et al., 2026; Fisman et al., 2025) and finance (Drott et al., 2024; Avdeenko et al., 2026).
6. Bridge to the **Colombia-Venezuela exercise**.

Why firm-level matters



Why effects show up at home

Standard macro intuition: sanctions are a tax on trade with the target. Why should firms in the *sanctioning* country be hurt beyond the standard welfare triangle?

- **Weaponized prior dependence** — existing relationships embody sunk costs (Eaton et al., 2011).
→ a firm that exits Russia does not just lose a customer, it loses a search investment.
- **Costlier alternatives** — substitution is partial; some products have no near substitute.
- **Fewer varieties** on the sender side — imports of Russian inputs (gas, metals, fertilisers) re-priced.

Source: IATRC keynote frame 7.

Four channels for self-inflicted damage

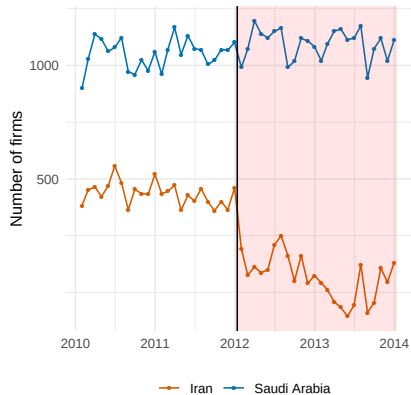
Beyond the formal embargo, firms face four overlapping channels:

- **Consumer/investor boycott** — reputational pressure (Hart et al., 2024; Biermann and Leromain, 2025).
- **Trade finance** — letters of credit, correspondent banking, insurance (Crozet and Hinz, 2020; Drott et al., 2024).
- **Country risk** — ruble volatility, war risk, contract enforceability (Biermann and Leromain, 2025).
- **Compliance cost** — legal review, dual-use screening, end-user certificates.

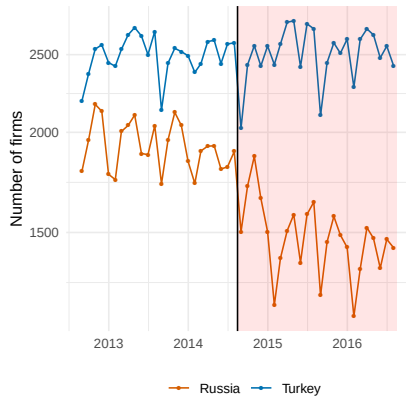
Source: IATRC keynote frames 7–8 and Crozet et al. (2025), §4.

**Extensive margin in the
sanctioning country**

Number of French exporters — Iran 2012 & Russia 2014



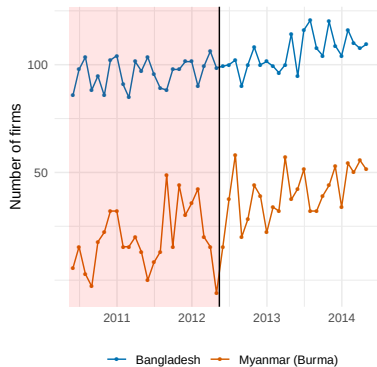
Iran sanctions tightened mid-2012



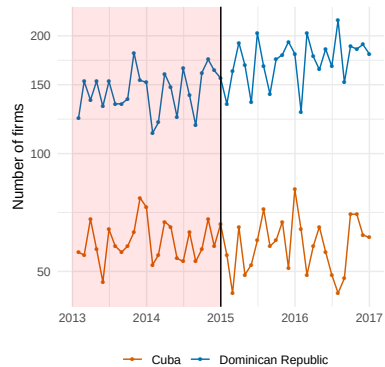
Russia sanctions July 2014

Source: Crozet et al. (2021), French customs (SIREN $\times$ NC8 $\times$ dest. $\times$ month).

Lifting sanctions is *not* symmetric — Myanmar vs. Cuba



Myanmar: +1/3 exporters after 2012 lifting



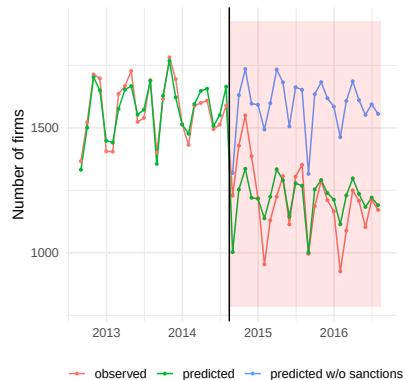
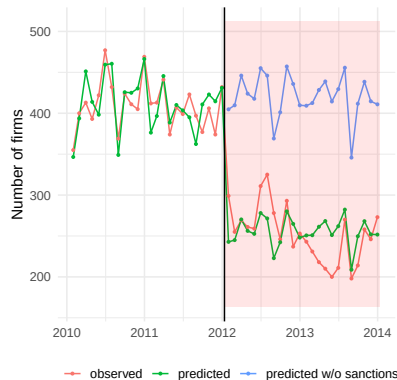
Cuba: essentially flat after Obama 2014 thaw

Source: Crozet et al. (2021). The **asymmetry** motivates the Colombia–Venezuela design.

Crozet, Hinz, Stammann & Wanner (2021) — research design

- Setting — monthly French customs 2009–2016, $\approx$ 150k firms, $>$ 200 destinations.
- Question — once sanctioned, do firms *exit* or just *shrink*?
- Design — 3-way fixed-effects **probit** on the entry/continuation indicator, with **bias correction** (Crozet et al., 2021).
- Three specifications:
 - baseline — $\Pr(X_{fdt} > 0) = \Phi(\delta \text{Sanction}_{dt} + \theta_f + \theta_d + \theta_t)$.
 - with prior experience — interact with lagged exporting indicator.
 - with firm characteristics — size, productivity, trade-finance intensity.

Counterfactual number of exporters

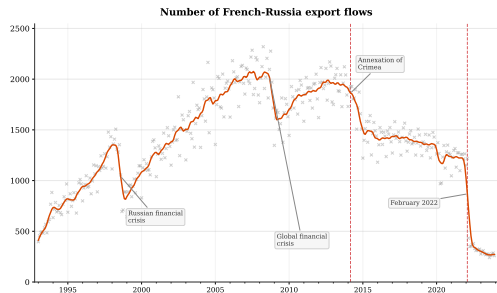


Iran: —40% exporters in counterfactual peak. Russia: —23%. Firms that exit **do not return**. *Source:* Crozet et al. (2021).

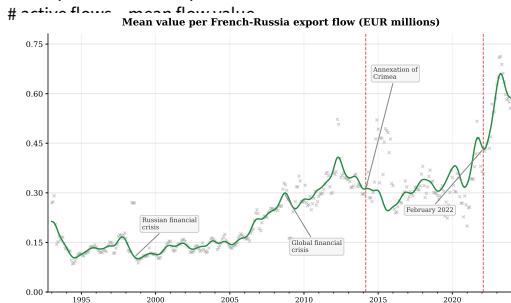
Estimating margins of adjustment

Frame 4a — trade as X_{ij} : the margin decomposition

Aggregate bilateral exports decompose as $X_{ij,t} = \underbrace{N_{ij,t}}_{\text{\# active flows}} \cdot \underbrace{\bar{v}_{ij,t}}_{\text{mean flow value}}$



$N_{ij,t}$ — number of French export flows to Russia



$\bar{v}_{ij,t}$ — mean value per flow

2022: N collapses ($1200 \rightarrow 300$), $\bar{v}$ rises (composition). **The extensive margin does the heavy lifting.** Source: Crozet et al. (2025), §5, Figs. 9b–c.

Frame 4b — PPML for the intensive margin

Gravity in multiplicative form:

$$X_{fpdt} = \exp(\beta \text{Treat}_{dt} + \theta_{fpd} + \theta_{fpt}) \cdot \varepsilon_{fpdt}.$$

- **Why PPML over log-OLS?** (Santos Silva and Tenreyro, 2006)
 - log-OLS drops zeros; PPML uses the full panel.
 - log-OLS estimates of β are biased under heteroskedasticity ($\text{Var}(\varepsilon \mid \cdot)$ depends on covariates).
 - PPML uses the moment $\mathbb{E}[X \mid \cdot] = \exp(\cdot)$ — consistent under standard regularity.
- **Conditional-mean interpretation:** $\hat{\beta}$ is the elasticity of the *expected value* of X_{ij} , *not* of $\log X_{ij}$ (Santos Silva and Tenreyro, 2006). The -74.6% on frame 4e is $\exp(\hat{\beta}) - 1$ from $\hat{\beta} = -1.369$.

Frame 4b — PPML for the intensive margin (cont.)

- **Fixed-effect choice** (review-paper §5.1):
 - firm $\times$ product $\times$ destination (θ_{fpd}) — absorbs the gravity-pair-product baseline.
 - firm $\times$ product $\times$ time (θ_{fpt}) — absorbs aggregate firm-product shocks.
- **Relation to 4d:** the PPML structure and the triple-diff on frame 4d are *nested*, not parallel — the PPML becomes the triple-diff once treatment varies on the product dimension (destination $\times$ time $\times$ sanctioned-product status), distinguishing sanctioned from non-sanctioned HS6 codes within (d, t) .
- Estimator — `fixest::fepois()` in R; the Colombia exercise uses exactly this command.

Frame 4c — LPM / probit for the extensive margin

Define the entry/continuation indicator $y_{fpdt} = \mathbf{1}\{X_{fpdt} > 0\}$. Two routes:

- **Linear probability model** (LPM):

$$y_{fpdt} = \gamma \text{Treat}_{dt} + \theta_{fpd} + \theta_{fpt} + u_{fpdt}.$$

Fast, handles high-dimensional FE; coefficients are average partial effects.

- **3-way fixed-effects probit** (Crozet et al., 2021): incidental-parameter bias is *not* negligible with three FE dimensions.

→ **Incidental-parameters problem** (Neyman and Scott, 1948): nonlinear ML estimators with high-dimensional FE are biased of order $O(1/T)$. The Stammann (2019) / Fernández-Val and Weidner (2016) **analytical correction** subtracts the leading bias term and restores consistency.

Frame 4c — LPM / probit for the extensive margin (cont.)

- **Software, honestly:**

- In R, the bias-corrected probit lives in `alpaca::feglm()` — `fixest`'s analytical bias correction is implemented for *Poisson* only, not for probit.

- The Colombia exercise uses `fixest::femlm` for PPML (works) and *falls back to plain LPM* for the extensive margin (no bias correction needed because LPM is linear).

- Trade-off: LPM is robust and transparent; probit handles non-linearities and is what Crozet et al. (2021) use to identify entry costs.

Frame 4c.5 — the (non-trivial) balanced-panel step

The students-always-ask question: **what about the zeros?**

- PPML on the intensive margin (frame 4b) is run on the *sparse* flow panel — one row per observed positive shipment. PPML eats zeros where they exist, but does not invent them.
- The LPM (and probit) on $\mathbf{1}\{X_{fpdt} > 0\}$ needs the **balanced** firm $\times$ product $\times$ destination $\times$ year frame — every potential cell, with 0 where no shipment was observed.
- In R: `data.table::CJ(firms_products, destinations, years)` cross-joins the dimensions. On Colombian DIAN this gives $\approx 150\text{k firms} \times \text{thousands of HS6 codes} \times 35 \text{ destinations} \times 10 \text{ years}$ — typically **10–20M rows**.
- Pass `mem.clean = TRUE` to `fixest::feols/femlm`: clears the auxiliary objects between FE projections so the regression fits in 16 GB of RAM.
- Pedagogical point: a naive LPM run on the *flow* panel (positives only) is wrong — the dependent variable is then identically 1.

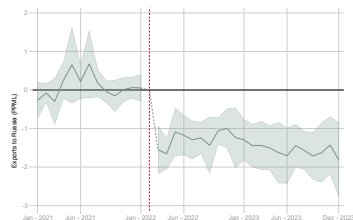
Frame 4d — triple-difference identification

Review paper (Crozet et al., 2025) §5.1 specification:

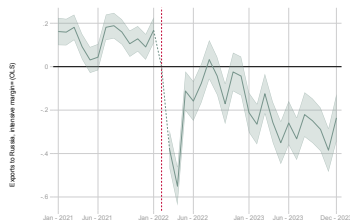
$$Trade_{ikdt} = \alpha Treatment_{dt} + \underbrace{\theta_{ikt}}_{\text{firm} \times \text{product} \times \text{time}} + \underbrace{\theta_{ikd}}_{\text{firm} \times \text{product} \times \text{destination}} + \varepsilon_{ikdt}$$

- θ_{ikt} absorbs the **firm's time-varying global demand for product k** — related to, but distinct from, country-level outward multilateral resistance.
- θ_{ikd} absorbs **firm-product-pair gravity** — structural distance / preference between firm-product and destination.
- $Treatment_{dt}$ **varies in three dimensions**: destination, time, and (interacted with sanctioned-product status) HS code — hence “**triple**”-diff.
- Identification — compares treated (d, t) cells to other destinations *within the same firm-product-time*.
- Run as PPML on total value, OLS on $\log(X)$ for intensive, LPM on $\mathbf{1}\{X > 0\}$ for extensive.

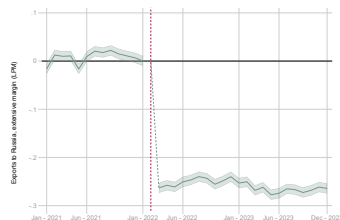
Frame 4e — bringing it together: French triple-diff, 2022 Russia



PPML — total (−74.6%)



OLS log — intensive (−25.4%)



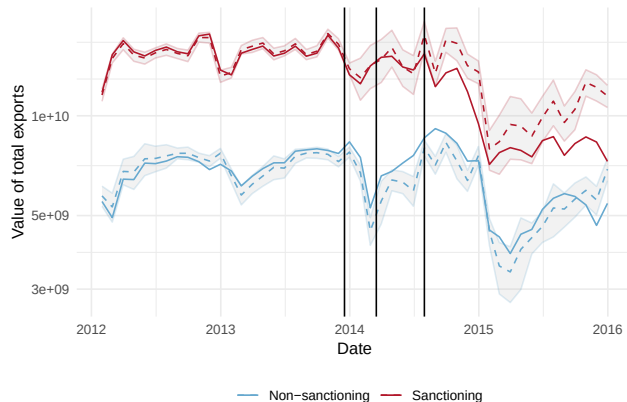
LPM — extensive (−75.9% of baseline 0.34)

- Flat pre-trends; sharp, persistent break at Feb 2022; extensive margin does *not* recover.
- **Handoff:** after the break we replicate this on **Colombian** customs around the 2014–2017 Venezuela collapse — DIAN microdata, PPML + LPM, placebos to Ecuador / Peru / Mexico.

Source: Crozet et al. (2025), Figs. 12a–c.

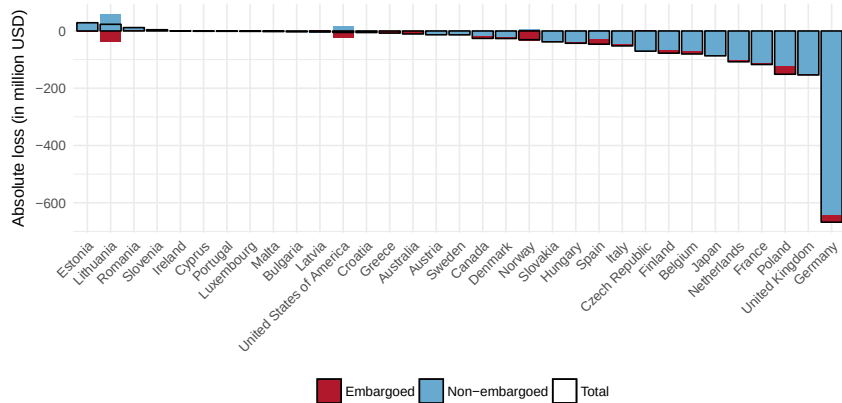
Intensive margin: Friendly Fire

Crozet & Hinz (2020) — predicted vs. observed



Gravity counterfactual: post-2014 sender exports fall **far below** the predicted baseline; most of the gap is in *non-embargoed goods*. *Source:* Crozet and Hinz (2020), monthly gravity, treatment window 2014m8 onwards.

Lost trade by sender — the bar chart



Average monthly export loss (USD m). Germany and the Netherlands lead in absolute terms; small open economies (Lithuania, Finland) lead in share of bilateral exports. *Source: Crozet and Hinz (2020).*

Which channel? — trade-finance interaction

Loss is concentrated in *non-embargoed* goods $\Rightarrow$ the channel is **not the embargo per se** but financial / risk frictions.

Outcome (PPML on flow value)	Products	Firms
Treatment $\times$ post-2014	−0.41***	−0.37***
Treatment $\times$ post-2014 $\times$ LC-share	−0.62***	−0.58***
Firm $\times$ product $\times$ destination FE	yes	yes
Firm $\times$ product $\times$ time FE	yes	yes

Coefficients on the triple interaction are **more negative** for goods/firms with high letter-of-credit reliance — consistent with the **trade-finance channel** (Crozet and Hinz, 2020; Drott et al., 2024).

Source: Stylised version of IATRC frame 21 / Crozet and Hinz (2020), Table 5.

Effects on the target

Aytun, Hinz & Özgüzel — the Su-24 embargo

Nov 2015: a Turkish F-16 downs a Russian Su-24. From 1 Jan 2016 Russia embargoes **17 Turkish products** (mostly fresh produce); lifted product-by-product through 2017.



Three-way FE PPML: **embargo** ≈ -13 log points
on flow value; diversion / circumvention rows
track whether lost exports re-appear elsewhere.

	Two-way	Two-way with est. FE	Three-way	Three-way with global data
Embargo x period imposition	-14.36*** (0.6491)			
Embargo x period lifting	-0.7000*** (0.0837)			
Diversion x period imposition	0.0831 (0.0529)			
Diversion x period lifting	-0.2351*** (0.0567)			
Circumvention x period imposition	-0.4435*** (0.0520)			
Circumvention x period lifting	-0.2090*** (0.0537)			
Est. destination $\times$ product $\times$ time FE				
Observations	1,185,212	1,114,179	1,179,861	13,085,742
Origin $\times$ product $\times$ time FE	yes	yes	yes	yes
Origin $\times$ destination $\times$ product $\times$ month FE	yes	yes	yes	yes
Destination $\times$ time FE	no	no	yes	no
Destination $\times$ product $\times$ time FE	no	no	no	yes

Source: Aytun et al. (2025); IATRC keynote frames 27–31.

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track whether lost exports re-appear elsewhere.

	Two-way	Two-way with est. FE	Three-way	Three-way with global data
Embargo x period imposition	-14.36*** (0.6491)	-12.93*** (0.7108)		
Embargo x period lifting	-0.7000*** (0.0837)	-0.1752** (0.0672)		
Diversion x period imposition	0.0831 (0.0529)	0.1729*** (0.0478)		
Diversion x period lifting	-0.2351*** (0.0567)	-0.0488 (0.0450)		
Circumvention x period imposition	-0.4435*** (0.0520)	-0.4077*** (0.0471)		
Circumvention x period lifting	-0.2090*** (0.0537)	-0.1058* (0.0476)		
Est. destination $\times$ product $\times$ time FE		0.8567*** (0.0161)		
Observations	1,185,212	1,114,179	1,179,861	13,085,742
Origin $\times$ product $\times$ time FE	yes	yes	yes	yes
Origin $\times$ destination $\times$ product $\times$ month FE	yes	yes	yes	yes
Destination $\times$ time FE	no	no	yes	no
Destination $\times$ product $\times$ time FE	no	no	no	yes

Source: Aytun et al. (2025); IATRC keynote frames 27–31.

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Three-way FE PPML: **embargo $\approx$ -13 log points**

on flow value; diversion / circumvention rows

track whether lost exports re-appear elsewhere.

	Two-way	Two-way with est. FE	Three-way	Three-way with global data
Embargo x period imposition	-14.36*** (0.6491)	-12.93*** (0.7108)	-13.62*** (1.042)	
Embargo x period lifting	-0.7000*** (0.0837)	-0.1752** (0.0672)	-0.1518 (0.0981)	
Diversion x period imposition	0.0831 (0.0529)	0.1729*** (0.0478)	0.0607 (0.0683)	
Diversion x period lifting	-0.2351*** (0.0567)	-0.0488 (0.0450)	-0.0977 (0.0653)	
Circumvention x period imposition	-0.4435*** (0.0520)	-0.4077*** (0.0471)	0.0813 (0.0832)	
Circumvention x period lifting	-0.2090*** (0.0537)	-0.1058* (0.0476)	0.5076*** (0.0887)	
Est. destination $\times$ product $\times$ time FE		0.8567*** (0.0161)		
Observations	1,185,212	1,114,179	1,179,861	13,085,742
Origin $\times$ product $\times$ time FE	yes	yes	yes	yes
Origin $\times$ destination $\times$ product $\times$ month FE	yes	yes	yes	yes
Destination $\times$ time FE	no	no	yes	no
Destination $\times$ product $\times$ time FE	no	no	no	yes

Source: Aytun et al. (2025); IATRC keynote frames 27–31.

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Three-way FE PPML: **embargo ≈ -13 log points**

on flow value; diversion / circumvention rows

track whether lost exports re-appear elsewhere.

	Two-way	Two-way with est. FE	Three-way	Three-way with global data
Embargo x period imposition	-14.36*** (0.6491)	-12.93*** (0.7108)	-13.62*** (1.042)	-13.05*** (0.6618)
Embargo x period lifting	-0.7000*** (0.0837)	-0.1752** (0.0672)	-0.1518 (0.0981)	-0.2994** (0.1024)
Diversion x period imposition	0.0831 (0.0529)	0.1729*** (0.0478)	0.0607 (0.0683)	0.6815*** (0.0807)
Diversion x period lifting	-0.2351*** (0.0567)	-0.0488 (0.0450)	-0.0977 (0.0653)	0.2474*** (0.0717)
Circumvention x period imposition	-0.4435*** (0.0520)	-0.4077*** (0.0471)	0.0813 (0.0832)	-0.0090 (0.0982)
Circumvention x period lifting	-0.2090*** (0.0537)	-0.1058* (0.0476)	0.5076*** (0.0887)	0.1572 (0.1030)
Est. destination $\times$ product $\times$ time FE		0.8567*** (0.0161)		
Observations	1,185,212	1,114,179	1,179,861	13,085,742
Origin $\times$ product $\times$ time FE	yes	yes	yes	yes
Origin $\times$ destination $\times$ product $\times$ month FE	yes	yes	yes	yes
Destination $\times$ time FE	no	no	yes	no
Destination $\times$ product $\times$ time FE	no	no	no	yes

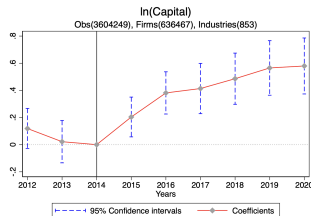
Source: Aytun et al. (2025); IATRC keynote frames 27–31.

Embargo bites beyond exports — employment at home

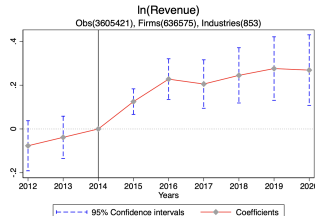
Dependent Variable:	log(total workers)		
Model:	(1)	(2)	(3)
Embargo × period imposition	-0.1085** (0.0419)	-0.0989** (0.0425)	-0.1108** (0.0426)
Embargo × period lifting	-0.1161** (0.0498)	-0.1159** (0.0498)	-0.1132** (0.0510)
Non-Russia × period imposition		0.0522 (0.0460)	
Non-Russia × period lifting		0.0298 (0.0500)	
Non-embargo × period imposition			-0.0114 (0.0486)
Non-embargo × period lifting			0.0230 (0.0602)
Observations	88,553	88,553	88,553
R ²	0.92214	0.92216	0.92215

Treated Turkish firms lose $\approx 11\%$ of monthly establishment employment during imposition; the effect **persists** after lifting. *Source:* Aytun et al. (2025), employment regressions.

Nigmatulina — “smart” sanctions backfire on the target



Capital of sanctioned firms



Revenue of sanctioned firms

- 600k Russian firm balance sheets, 2014–2020. Targeted firms gain +32% capital, +20% revenue as state procurement & subsidies are redirected to them; aggregate TFP loss $\approx 1.1\%$ (Nigmatulina, 2022).
- Compare: Miromanova (2023) (Russia 2014, ext. + int. bind, unit values $\uparrow$); Görg et al. (2024) (Germany 2022, heterogeneous by size); Ahn and Ludema (2020) (Russia 2014 target side, -25% rev., -50% assets, strategic firms shielded).

Frontier: circumvention and finance

Scheckenhofer, Teti & Wanner (2025) — military goods (+20 pp)

- Data — UN Comtrade HS6 annual, 2021–2023; 44 HS6 “Common High-Priority” military codes (EU/G7 Oct 2023 list).
- Two triple-DiDs:
 - $\Pr(\text{export to RUS}) = \beta_1 \text{ after}_t \cdot \text{friends}_i \cdot \text{RUS}_j + \text{FEs} \Rightarrow \hat{\beta}_1 = +20 \text{ pp.}$
 - $\Pr(\text{export to friends}) = \beta_2 \text{ after}_t \cdot \text{allies}_i \cdot \text{friends}_j + \text{FEs} \Rightarrow +4 \text{ pp.}$
- “Friends” — AM, AZ, CN, GE, KZ, KG, MD, RS, TR, UZ; FE structure absorbs general diversion.
- Re-export DiD (Georgia / Moldova / Uzbekistan): +7.6 pp evasion of military goods to Russia.

Headline policy take: **enforcement is the binding constraint**, not the bans themselves. *Source:*

Scheckenhofer et al. (2025), CESifo WP 11743 / AEA P&P 2025.

Eurasian Roundabout & the Undoing of Sanctions

- **Eurasian Roundabout** (Chupilkin et al., 2026)
 - bill-of-lading + customs from Caucasus and Central Asia; sanctioned-good imports from EU/G7 *into* AM/KZ/KG up 100–200%.
 - same products re-appear in those countries' exports *to* Russia; price wedge +20–40%.
- **The Undoing of Economic Sanctions** (Fisman et al., 2025)
 - conflict-zone trade and platform-mediated illicit rechanneling; sanctions get *undone* rather than evaded openly.
- Common point — the relevant counterfactual is not “no trade” but “traded through an intermediary, at a markup”.

Financial sleeve — TARGET2 and online platforms

- Drott et al. (2024) — TARGET2 microdata, Russian banks
 - SWIFT exclusion shows up as a sharp, lasting break in payment flows of designated banks.
 - spillovers: non-designated Russian banks rerouted via designated correspondents.
- Avdeenko et al. (2026) — online intermediaries on platforms
 - sanctioned-country sellers and buyers maintain commercial presence via marketplace pivots.
 - stigma raises prices but does not kill the channel; cross-border listings shift to neutral hubs.
- Berthou (2023) — sanctions risk slowly tilts **invoicing currencies** away from the dollar.

Bridge to Exercise 1

Colombia-Venezuela — data architecture

- Setting — 2014–2017 **collapse of Venezuelan demand**: a sudden, exogenous (to Colombian firms) shut-down of one of the two largest export markets.
- **Important disclaimer**: the 2014–2017 Venezuela episode is a *demand collapse + foreign-exchange-control regime*, not a Western sanction. We use it as a sanction analogue because (a) the firm-level response margins are identical, and (b) Colombian DIAN is the cleanest customs micro-panel publicly available.
- Data — DIAN customs, raw monthly $\text{NIT} \times \text{HS6} \times \text{destination} \times \text{value}$; we **aggregate to the firm-HS6-destination-year level** for tractability (monthly $\times$ balanced LPM panel blows up RAM).
- Pre-shipped: 00-setup.R, 01-download.R, 02-clean.R. Students start from a clean panel.

Colombia-Venezuela — data architecture (cont.)

- Treatment — Venezuela $\times$ post-2014; placebo destinations Ecuador, Peru, Mexico, USA.
- Triple-diff at the *firm-HS6* level (review-paper §5.1, mapped to the available FE dimensions):

$$X_{fpdt} = \alpha \text{Treat}_{dt} + \theta_{fp} + \theta_{ft} + \theta_{pdt} + \varepsilon_{fpdt}.$$

- Firm-product, firm-year, and product-destination-year FEs — identification off within-firm variation across destinations *and* sanctioned vs. placebo products.

What you will run — 03-estimate.R + 04-output.R

- **PPML for the intensive margin** (Santos Silva and Tenreyro, 2006):

```
fepois(value_usd ~ sanction | nit^hs6 + nit^year + hs6^destination^year,  
       data)
```

- **LPM for the extensive margin** (run on the *balanced* firm $\times$ HS6 $\times$ dest $\times$ year panel built via `CJ()`; `passmem.clean = TRUE`):

```
feols(exporting ~ sanction | nit^hs6 + nit^year + hs6^destination^year,  
      balanced)
```

- **Probit with bias correction** (Crozet et al., 2021) — shown for reference; *not* run in class. The exercise stops at LPM (see frame 4c on alpaca).
- Event-study plot, margins decomposition, comparison table. Compare to Crozet and Hinz (2020) and Aytun et al. (2025).

```
cd exercises/01-colombia-venezuela && make
```

Take-aways

Take-aways — Part 2

- Sanctions act **mostly through the extensive margin**, and the exit is **persistent** (Crozet and Hinz, 2020; Crozet et al., 2021).
- Costs accrue at home through **trade finance, country risk, and private compliance**, not just the formal embargo.
- On the target side, **“smart” sanctions get diluted** by state shielding (Nigmatulina, 2022; Ahn and Ludema, 2020); embargo episodes hit **employment** (Aytun et al., 2025; Görg et al., 2024).
- The 2022 frontier is **enforcement**: military-goods evasion +20 pp (Scheckenhofer et al., 2025); Eurasian roundabout (Chupilkin et al., 2026); finance (Drott et al., 2024).
- Methods to take home: **PPML + LPM + 3-way FE** with bias correction; **triple-diff** with θ_{fpd} and θ_{fpt} .

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